INT201 Decision, Computation and Language

Tutorial 4

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review

1. Regular language closed under operations.

definition RL : 1. can be recognize by DFA.

2. if and only if can be accepted by NFA

5 operation
1. union
2. concatination
3. kleene star
4. complement.
5. intersection -

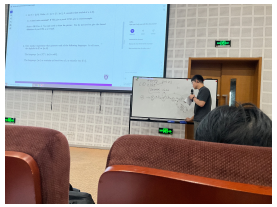
Proof:
1. math way -
2. Graph.
   1. construct new 5 tuple
   2. draw new Graph.

$\Rightarrow$ Constructive Proof.

2. Regular Expression

要考 !!!

1. Let $\Sigma = \{a, b\}$. Define $A = \{\, w \in \Sigma^* \mid |w| \geq 3,$ second-to-last symbol of w is $b \,\}$.



Is A closed under **reversal**? If YES, give a proof. If NO, give a counterexample.

Draw a DFA for A. You only need to draw the picture. You do not need to give the formal definition of your DFA as a 5-tuple.

Q. Σ, δ, q, F

NO, cause if we enter abb could be accepted by the original of DFA but the reversal bba can't be accepted cause the second-to-last must be symbol of w is b. but is a.

abb

reversal → bba

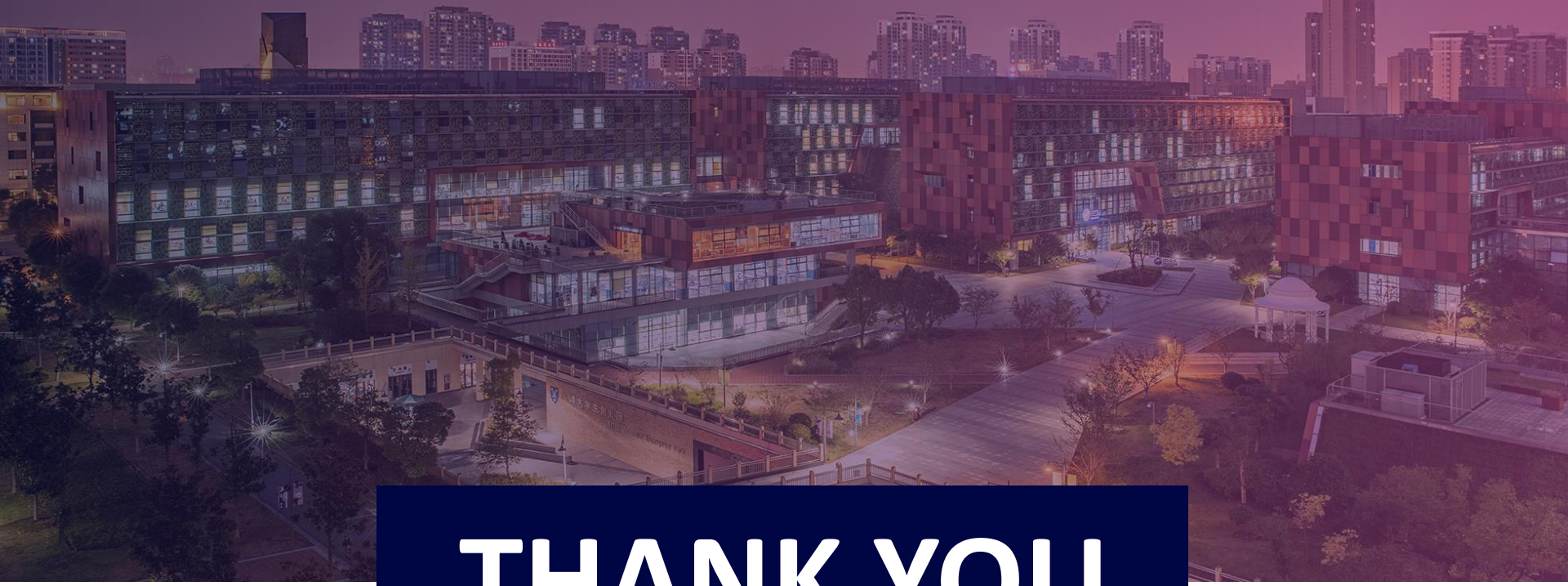
2. Give regular expressions that generate each of the following languages. In all cases, the alphabet is $\Sigma = \{a, b\}$.

The language $\{\, w \in \Sigma^* \mid |w| \text{ is odd} \,\}$. 1, 3, 5, 7 ...

The language $\{\, w \mid w$ contains at least two a's, or exactly two b's $\,\}$.

1. $(a \cup b)^*(bb \cup ba \cup aa \cup ab)^*$. $(a \cup b)((a \cup b)(a \cup b))^*$ + even => odd.

2. $aa(a \cup b)^* \cup bb(a)^*$. $b^* a b^* a (a \cup b)^* \cup a^* b a^* b a^*$



THANK YOU

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